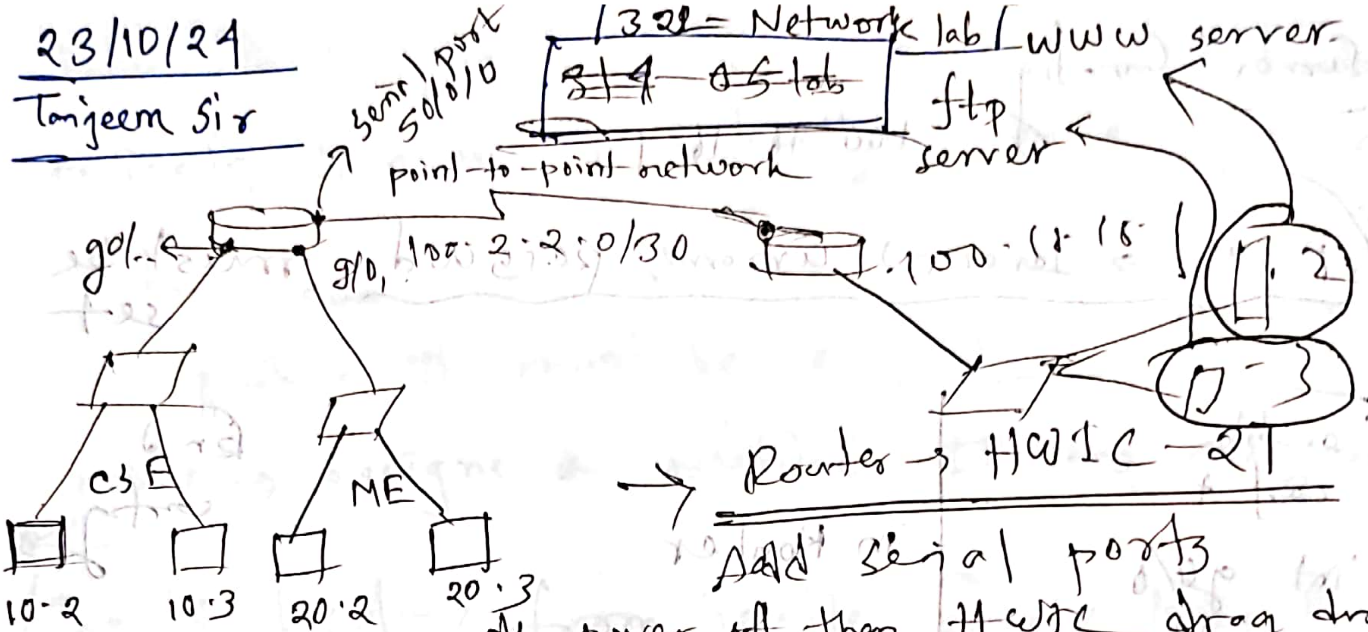
23/10/24

Tanjeem Sir



→ Router → HWIC-2T

Add serial ports

* power off then HWIC drag drop

192.168.10.0/24   192.168.20.0/24

Total $\boxed{2^7 - 2}$ hosts

↳ subnet mask net^k bits

All 0s → net^k

All 1s → broadcast addr

if only 2 hosts needed → then net^k bits = 30

/30 → 00, 11 — usable → 01, 10

00 ↳ net^k, 11 ↳ broadcast, 01, 10 ↳ hosts

→ End devices — ip, sub mask, def gateway must be set up

*** In router, first the ports will not come. We need to first power off, then drag and drop HWIC-2T option to bring ports.

Server Config

100.18.18.1

FTP → Server → uname, password must be set

enable
conf t
int g0/0
ip address ...
no shutdown

in Router

Serial DCE → One has clock rate

100.10.10.1

so one port of router will get the clock

int s0/0/0
ip address 100.2.2.2 255.255.255.252

subnet mask

Static Routing (default)

ip route 0.0.0.0 0.0.0.0 s0/0/0

matches all packets

Public IPs are limited → so NAT is needed
private IPs are just locally unique

Importance of NAT

when comm with internet → uniquely identified
public IP must be bought

NAT → assigns a public IP to everyone

Inside local → from inside, source looks like..
" Global → from outside, source looks like..
→ will be a public IP in our case

In this case → outside local, global same
public IP of server

Static NAT

Dynamic NAT
→ are from pool then
if more than public IP..

PAT → just assign port ~~inside~~ instead of ~~net~~ ip

→ practically impossible

ip nat inside source static

Setting static NAT — not much used

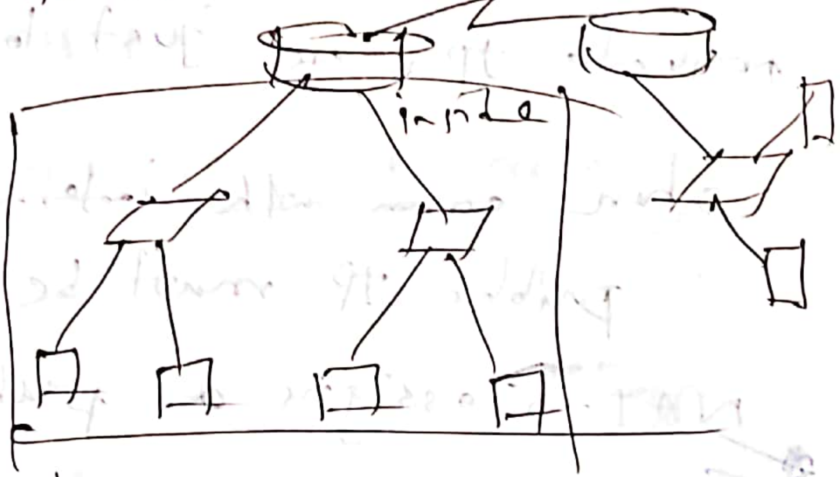
ip nat BUF POOL ..[209.175.200.5 ...]

setting a pool

access-list 1 permit

ACL deny maybe given too

↳ setting permissions



deny all → all others deny

show running-config | include ip nat

int g0/0
ip nat inside
→ all ints in this way.

show ip nat translations

first come first serve → public IPs set this way

NAT table is maintained → if prev existing → not again assigned

ping → single session

PAT

ip nat commands are same as nat

overload → just one line extra

same ip used → port different in PAT

Ping — ICMP

This allows more efficient public IP assignment.

End device

UDP may be used

range public IP ...

NAT, Dnat, PAT ...

Ping *

sh ip nat trans

from Desktop menu → HTML ...

removing permissions → ACL ...

sh ip nat statistics

which mode ...

clear ip nat trans *

→ this is needed for debugging, clearing prev logs

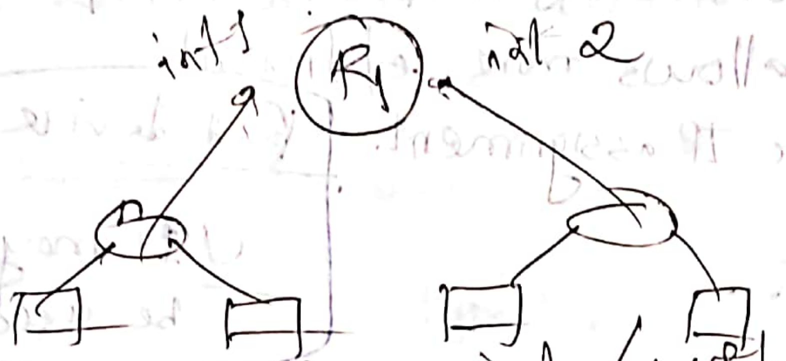
ACL - Access Control List

ACL statement

ACL entry

ACL two types
i) standard — src IP
ii) Extended — src, dst, port tcp, udp

* unnecessary ACLs make → time more comm slow as all packets are checked



inbound } wrt
outbound } Router

any port — inb / outb no problem

one p' interface ~~one~~ 4 ACLs

v4 inbounds
v4 outbound } → 4 ACLs ← { v6 inbound
v6 outbound

std → closest to dest

Std → only src can be mentioned
dest can't be filtered

closest to dest → commands given
otherwise it would block necessary traffic to the device

Ext → closest to src → filtering to

closest to src to filter

Standard, Extended ACL

List → number or name

0–99 → Std ← 1300–1399

In slide

ACL means a list

ACL
- ACE permit/deny
- ACE ...
- deny all

1st command first

2nd command cannot overturn the 1st command

ACL rules → sequence important

deny all → implicit

permit all is needed to override the implicit deny all

net mask 11 ... 1100..

wildcard mask → invert of subnet mask

net addr → wildcard mask

access-list 10 permit

192.168.20.0 0.0.0.255

any change has

not all 1's need to be consecutive

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Keep in mind that, commands given earlier have higher priority.

write to text file then apply

std → 1–99; 1300–1399

ext → 100–199,

access-list 10 permit host _______ only this IP

only this pc

access-list 10 deny 192.168.11.0 0.0.0.255 0.0.0.0

Examples:

ip addr	wildcard mask
10.2	0.0.0.1

2, 3 blocked

not bit change so, only 1 ip is affected

ip addr	wildcard mask
20.8	0.0.0.7

⇒ 20.8, → 20.15 blocked

0 0 0 1 0

ip access-group 10 out

1 0 0 0
1 1 1

How wild card mask works?

Say we have a IP A.B.C.D and wildcard mask P.Q.R.S

⇒ Now, it means, we can alter any of the bits in A.B.C.D that are set 1 in P.Q.R.S

So, 0.0.0.3 means we can freely alter the last two bits ip access-list standard permit access → setting by name

host ip

same os ip 0.0.0.0 → this means no bit can be offered, so only the one single IP

ftp

src IP src WC dest IP main

permit tcp

or www/80, http access

$\boxed{\text{DHCP}}$ → uses UDP → in this case
udp ... we need to write udp ftp/code

Ext ACL → slow and powerful service

permit ip any (src) any (dest) → if we miss this, ping won't work

permit any any

serial out serial

ARP req
in progress
failed

g0/0 in → only CSE

ip wildcard → any

Separately apply to Closest to src

otherwise many problem

Plan first

simulate

default route

— plan first
— and save checkpoints

| exact version | * Moodle

setup
www
ftp

Strategy → i) Plan first
ii) Write in notepad text
iii) ~~as~~ save checkpoints